# Example Project 2: Do Peer-to-Peer Borrowers Pay More Than Bank Borrowers?

EXAMPLE STUDENT REPORT (QUANTITATIVE VARIABLE), STAT 251 SECTION 01

> **INSTRUCTOR NOTE**
>
> This is a completed example of the course project. It shows what a strong report looks like when the variable being analyzed is **quantitative**. It follows the six steps on the Project Description (project.html) page in order, so you can compare it against the rubric section by section.
>
> The two example projects deliberately come from very different fields: Example Project 1 (example_project_categorical.html) uses weather records, and this one uses consumer lending. Any topic works as long as the data can be cited and your question fits a method from class. Use these examples to see the level of detail and explanation expected. **Do not copy their datasets, questions, or wording.**
>
> The example was prepared with the help of Claude (Anthropic), and every calculation in it was done with the course's stat-tools (https://stat-tools.jmks-homelab.dev/) site. The screenshots come from those tools. This example also shows that a well-run test that **fails to reject** $H_0$ is still a complete, strong project.

## 1. Dataset Selection

**(A) Citations.** I found the loan data on **Kaggle** and the benchmark on **FRED**. Both are data portals listed on the course project page.

> Singh, U. (2023). *Lending Club Loan Dataset* [Data set, file `loans_full_schema.csv` ]. Kaggle. License CC0: Public Domain. https://www.kaggle.com/datasets/utkarshx27/lending-club-loan-dataset. Accessed September 25, 2026.

> Original source of the same file: OpenIntro. *loans_full_schema: Loan data from Lending Club* [Data set]. https://www.openintro.org/data/index.php?data=loans_full_schema. The data were originally published by LendingClub.

> Board of Governors of the Federal Reserve System (US). *Finance Rate on Personal Loans at Commercial Banks, 24 Month Loan* [TERMCBPER24NS]. Retrieved from FRED, Federal Reserve Bank of St. Louis, https://fred.stlouisfed.org/series/TERMCBPER24NS. Accessed September 25, 2026.

I downloaded the Kaggle file and compared it with OpenIntro's copy at https://www.openintro.org/data/csv/loans_full_schema.csv. They contain the same 10,000 loans and the same values; Kaggle adds only a row-number column.

**(B) Background.**

- **The lender.** LendingClub is a *peer-to-peer* lending platform: borrowers apply online, and the loans are funded by individual and institutional investors rather than by a bank's deposits.
- **The rows and variables.** Each row of the file is **one loan that was actually issued**, not an application. There are **10,000 loans**, all issued from January to March 2018, with **55 variables** about the borrower and the loan. These include:
  - the borrower's income, employment and credit history
  - the loan amount, term (36 or 60 months) and purpose
  - the **interest rate**
  - a risk **grade** from A (lowest risk) to G (highest risk) that LendingClub assigned.
- **Who collected it.** LendingClub recorded these data as part of its business and published loan-level data for the public. OpenIntro prepared this 10,000-loan file for teaching, and a Kaggle user re-posted it.
- **How the 10,000 loans were chosen.** The documentation does not say, so I cannot claim they are a random sample of all LendingClub loans (see Limitations).
- **The benchmark (FRED).** The Federal Reserve reports the average rate commercial banks charge on 24-month personal loans in its *G.19 Consumer Credit* release. FRED lists it as a monthly series, but values are published only for February, May, August and November. The **February 2018** value, from the same quarter these loans were issued, is **10.22%**. I saved the full series as a CSV.

## 2. Question and Hypothesis Formulation

**(A) Question.** Peer-to-peer lenders advertise themselves as an alternative to banks. So did people who took out a 3-year LendingClub loan in early 2018 pay a **higher interest rate, on average, than the 10.22% banks were charging for personal loans** at the same time?

**(B) Variables.**

- **Interest rate** ( `interest_rate` , percent per year): **quantitative, continuous**, recorded to two decimal places. This is the variable I test: its mean is compared with the bank benchmark.
- **Grade** ( `grade` , A–G): **categorical, ordinal**. I do not test it, but I use it in Part 4 to explain the shape of the interest rates, because LendingClub sets the rate mainly from the grade.

**(C) Hypothesis in words.** I expect the mean interest rate on 36-month LendingClub loans to be **greater than** 10.22%. I chose the direction before looking at any data, because the question is whether peer-to-peer borrowers pay *more*.

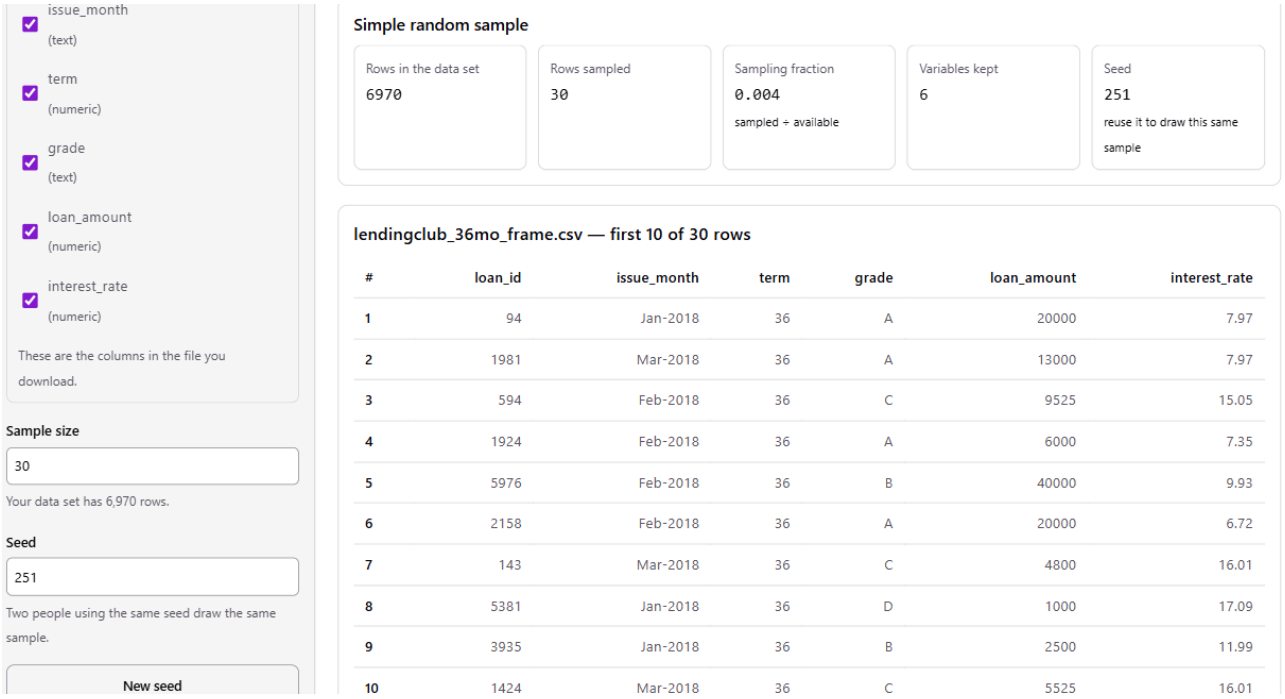
## 3. Sampling

**(A) and (B) Method.**

- **Population (sampling frame).** The **6,970 loans with a 36-month term**. I left out the 60-month loans because their term is even further from the benchmark's 24 months.
- **Sampling method.** I drew a **simple random sample of n = 30 loans** without replacement, using the stat-tools **Simple Random Sample** tool.

To repeat it:

1. Open `loans_full_schema.csv` in a spreadsheet. Add a column `loan_id` numbering the rows 1–10,000, keep only the rows with `term` = 36, and keep only the columns `loan_id` , `issue_month` , `term` , `grade` , `loan_amount` and `interest_rate` . Save it as a CSV (6,970 rows).
2. Open the **Simple Random Sample** tool and upload the CSV.
3. Keep all variables, set **Sample size = 30** and **Seed = 251**, and press *Draw sample*. The same seed always draws the same 30 loans.




| # | loan_id | issue_month | term | grade | loan_amount | interest_rate |
|---|---|---|---|---|---|---|
| 1 | 94 | Jan-2018 | 36 | A | 20000 | 7.97 |
| 2 | 1981 | Mar-2018 | 36 | A | 13000 | 7.97 |
| 3 | 594 | Feb-2018 | 36 | C | 9525 | 15.05 |
| 4 | 1924 | Feb-2018 | 36 | A | 6000 | 7.35 |
| 5 | 5976 | Feb-2018 | 36 | B | 40000 | 9.93 |
| 6 | 2158 | Feb-2018 | 36 | A | 20000 | 6.72 |
| 7 | 143 | Mar-2018 | 36 | C | 4800 | 16.01 |
| 8 | 5381 | Jan-2018 | 36 | D | 1000 | 17.09 |
| 9 | 3935 | Jan-2018 | 36 | B | 2500 | 11.99 |
| 10 | 1424 | Mar-2018 | 36 | C | 5525 | 16.01 |

**Figure 1.** The Simple Random Sample tool after the draw: 30 of 6,970 loans sampled with seed 251. The first rows of the sample are shown.

**(C) Justification.**

- **Why SRS.** Every 36-month loan has the same chance of being chosen, so the sample is not tilted toward any grade, month or loan size.
- **Why not stratify.** Stratifying by grade would guarantee every grade appears. But the question is about the *overall* mean rate, and an SRS already represents the grades in proportion to how common they are.
- **Independence.** The loans go to different borrowers, so they are close to independent. My sample is only 30 of 6,970 loans (0.4%), so sampling without replacement causes no problems.

**Possible bias.**

- **The sampling itself.** The SRS introduces none.
- **The file I sampled from.** The file is a 10,000-loan subset prepared by OpenIntro, and I do not know how those loans were chosen. If they are not representative of all LendingClub loans from early 2018, my results describe this file more than LendingClub as a whole.

**(D) The sample** is listed in full in Appendix A.

# 4. Exploratory Statistical Description

**(A) Descriptive statistics** from the stat-tools **Statistic Calculator** (R's default quartile definition):

**Table 1.** Descriptive statistics of the interest rate for the 30 sampled 36-month loans.

| Statistic | Value (%) |
|---|---|
| minimum | 5.3200 |
| Q1 | 7.3500 |
| median | 9.4400 |
| mean | 10.4187 |
| Q3 | 11.9900 |
| maximum | 26.3000 |
| interquartile range | 4.6400 |
| standard deviation | 4.5963 |

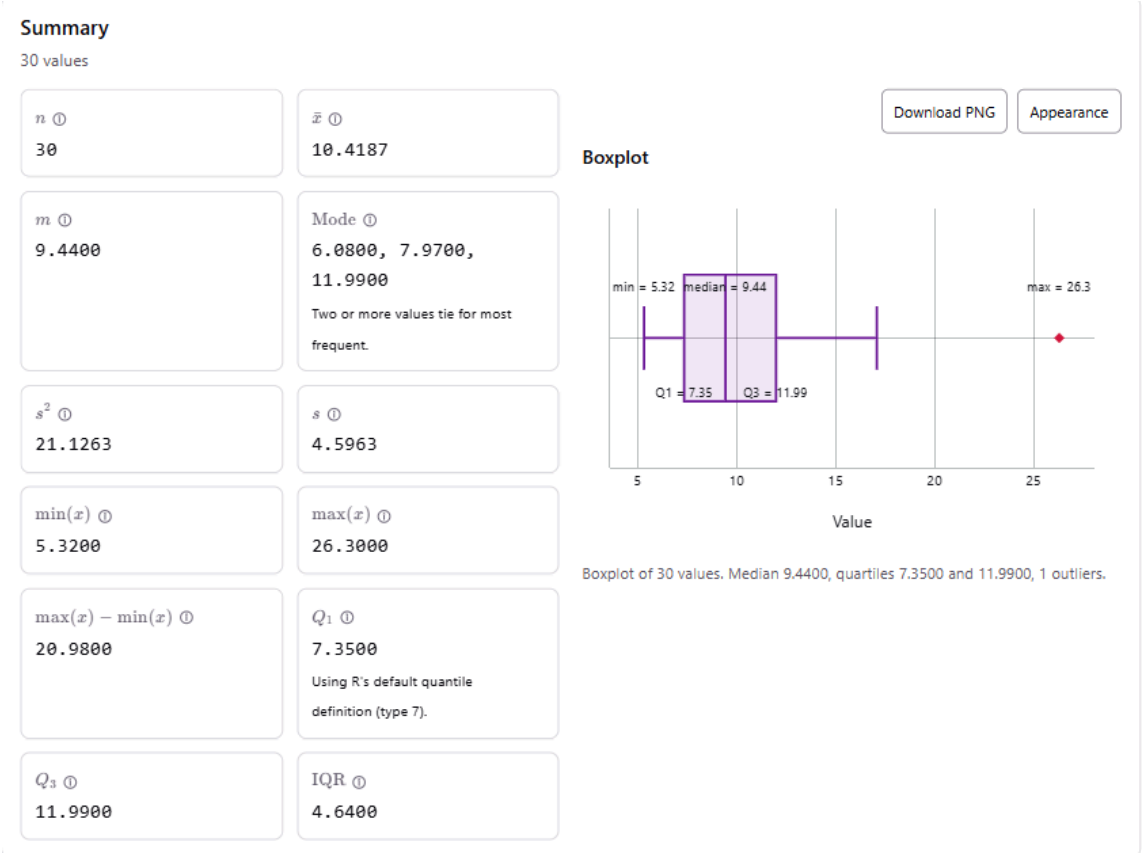



**Figure 2.** Statistic Calculator output and boxplot for the 30 sampled interest rates (%). The red point is the high outlier (26.30%).

**(B) Plots.** The boxplot is in Figure 2. The histogram below came from the **Make Graphs** tool, using its automatic bins:

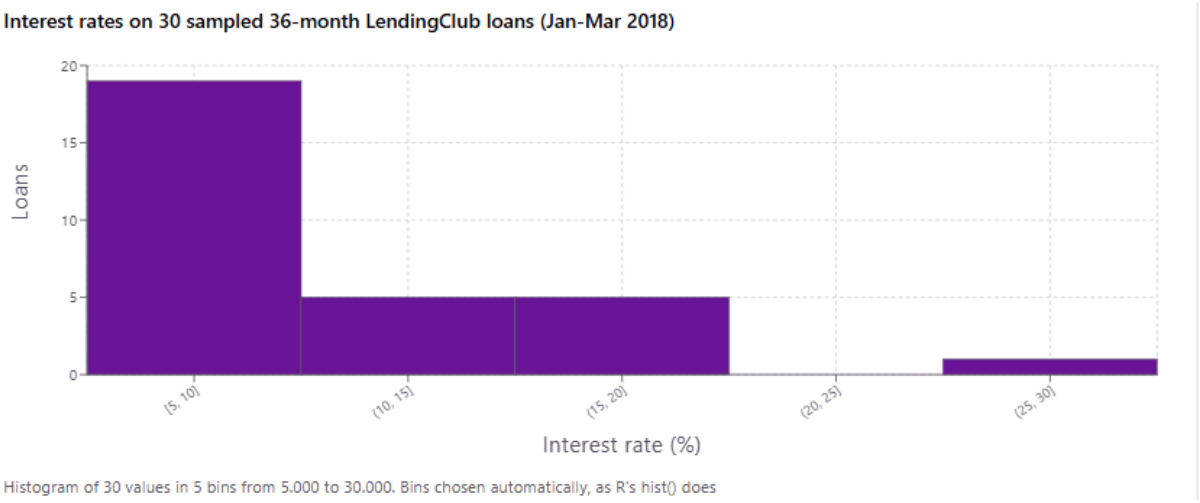



**Figure 3.** Histogram of the interest rate on 30 sampled 36-month LendingClub loans issued January–March 2018. The bins are 5 percentage points wide.

To understand the shape, I also counted the sampled loans by grade with the **Frequency Tables** tool:

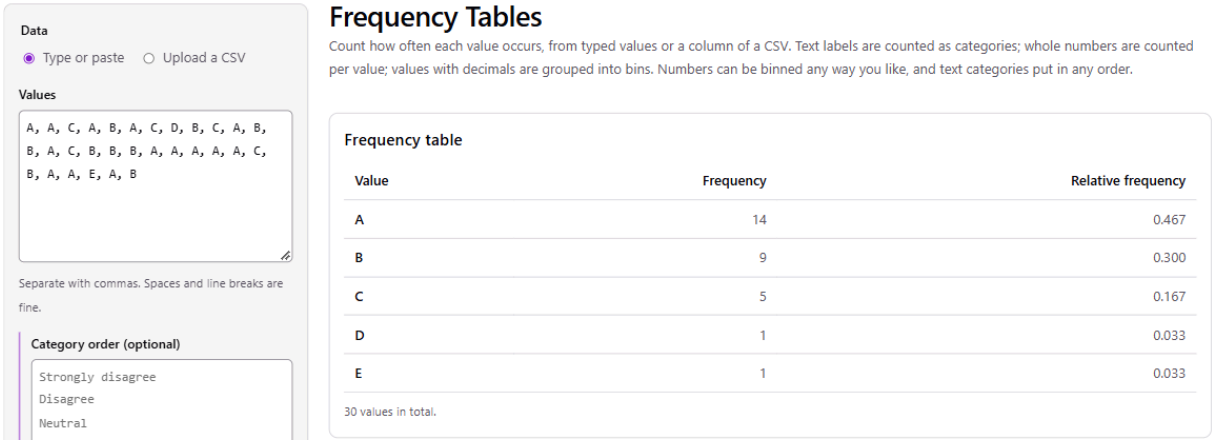



**Figure 4.** Frequency table of the LendingClub grade for the 30 sampled loans.

**Table 2.** Interest-rate range within each grade in my sample.

| Grade | Loans | Lowest rate (%) | Highest rate (%) |
|---|---|---|---|
| A | 14 | 5.32 | 7.97 |
| B | 9 | 9.44 | 11.99 |
| C | 5 | 13.58 | 16.02 |
| D | 1 | 17.09 | 17.09 |
| E | 1 | 26.30 | 26.30 |

**(C) Shape, center and variability.**

- **Shape.** The distribution is **clearly right-skewed**. Most loans (19 of 30) carry rates between 5% and 10%, and a few riskier loans stretch the tail out to 26%. The mean (10.42%) is about one percentage point above the median (9.44%), which fits the skew.
- **Why it is skewed.** Table 2 shows where the skew comes from. LendingClub sets rates by grade, and 23 of my 30 loans are grade A or B, the low-risk, low-rate grades. The few C, D and E loans are the ones that stretch the right tail.
- **Center.** The median rate is **9.44%** and the mean is **10.42%**. The benchmark (10.22%) falls between them.
- **Variability.** The standard deviation is **4.60 percentage points** and the IQR is **4.64 points**, so the middle half of the loans had rates from 7.35% to 11.99%.
- **Outliers.** The upper fence is $Q_3 + 1.5 \cdot IQR = 11.99 + 1.5(4.64) = 18.95\%$. The single grade-E loan at **26.30%** lies above it, so it is an outlier. It is a real loan, not an error: grade-E rates in the full file run in the mid-20s. I kept it.
    - The outlier pulls the mean *up*, which works in favour of my alternative hypothesis.
    - I check its influence in Part 5.

# 5. Hypothesis Testing

**(A) Hypotheses.** Let $\mu$ be the mean interest rate of 36-month LendingClub loans issued January–March 2018.

$$H_0 : \mu = 10.22\% \qquad\qquad H_a : \mu > 10.22\%$$

In words:

- $H_0$: peer-to-peer borrowers paid the same average rate as the bank benchmark.
- $H_a$: they paid more on average.

I use a significance level of $\alpha = 0.05$.

**(B) Choice of test and assumptions.** I use a **one-sample $t$ test for a mean**. I have one quantitative variable and want to compare its mean with a fixed benchmark. The population standard deviation is unknown, so the test uses $s$ and the $t$ distribution.

The assumptions are:

1. **Random sample.** Satisfied, because the loans were chosen by SRS (Part 3).
2. **Independent observations.** Satisfied: the loans go to different borrowers, and the sample is a tiny fraction of the frame.
3. **Approximately normal population, or a large enough sample.** This is the assumption to watch. The rates are right-skewed, with one outlier. With $n = 30$, the Central Limit Theorem makes the sampling distribution of $\bar{x}$ approximately normal for moderate skew like this, so the $t$ test is reasonable. I also rerun it without the outlier below.

**(C) Computations.**

*Step 1: observed statistics* (Table 1): $n = 30$, $\bar{x} = 10.4187\%$, $s = 4.5963\%$.

*Step 2: standard error.*

$$SE = \frac{s}{\sqrt{n}} = \frac{4.5963}{\sqrt{30}} = \frac{4.5963}{5.4772} = 0.8392$$

*Step 3: test statistic.*

$$t_{obs} = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{10.4187 - 10.22}{0.8392} = \frac{0.1987}{0.8392} = 0.237$$

*Step 4: degrees of freedom and critical value.* $df = n - 1 = 29$. For an upper-tail test at $\alpha = 0.05$ the critical value is $t_{29,\,0.95} = 1.699$, and we reject $H_0$ only if $t_{obs} \geq 1.699$.

*Step 5: p-value.* $p = P(t_{29} \geq 0.237) = 0.407$.

*Step 6: 95% confidence interval for $\mu$* ($t_{29,\,0.975} = 2.045$):

$$\bar{x} \pm t_{29,\,0.975}\, SE = 10.4187 \pm 2.045(0.8392) = 10.4187 \pm 1.716 = (8.70\%,\ 12.14\%)$$

I ran the test in the stat-tools **One Sample Tests** tool with these settings:

- Mean, hypothesis test, upper tail, $\alpha = 0.05$, $\mu_0 = 10.22$.
- Data: the sample CSV uploaded, with the `interest_rate` column chosen.

It reproduces every value above:

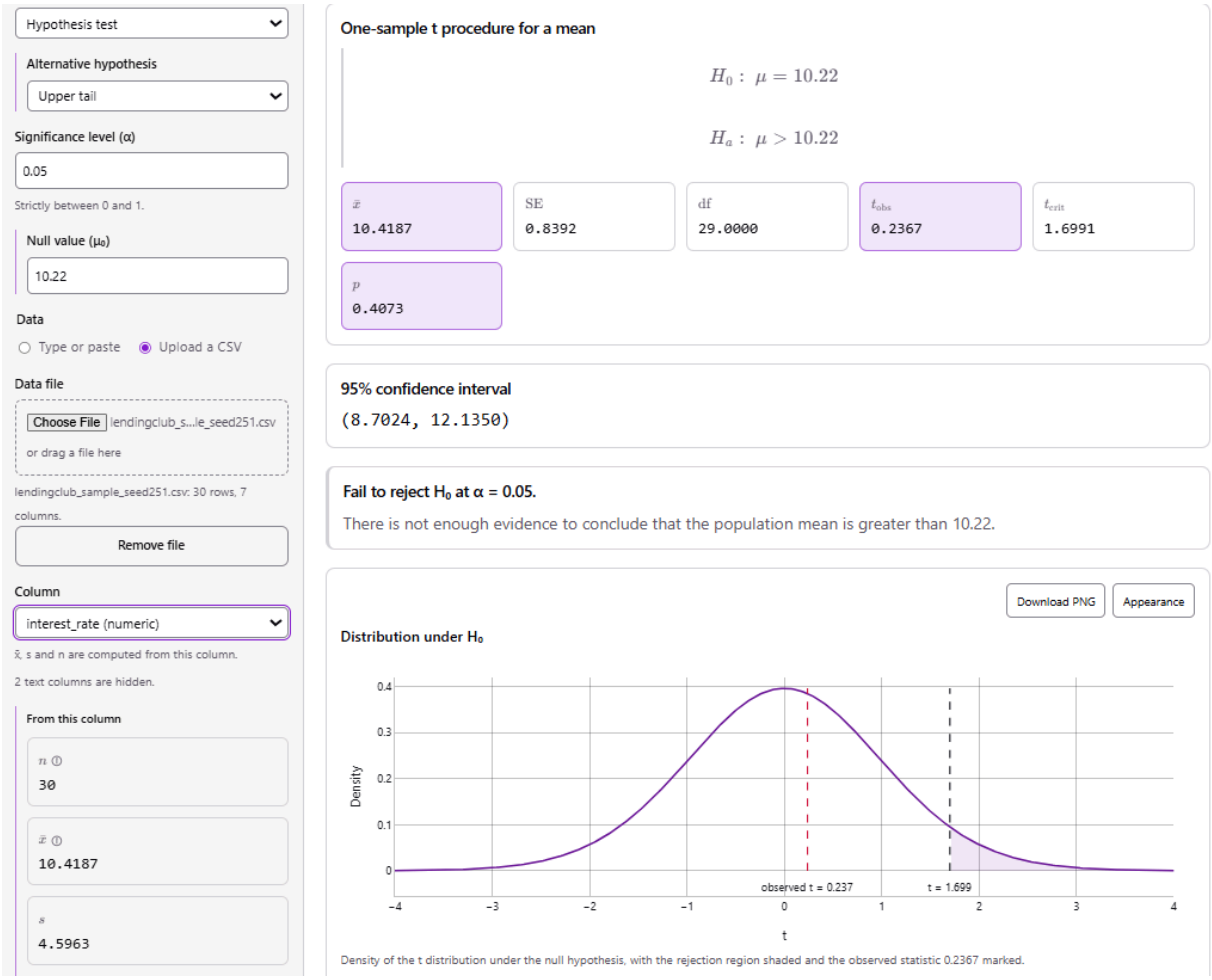



**Figure 5.** One Sample Tests tool output: SE = 0.8392, df = 29, t = 0.2367, critical value 1.6991, p = 0.4073, and 95% CI (8.7024, 12.1350). The observed t falls far short of the shaded rejection region.

| Quantity | Value |
|---|---|
| Sample mean $\bar{x}$ | 10.42% |
| Standard error | 0.839 |
| Degrees of freedom | 29 |
| Test statistic $t_{obs}$ | 0.237 |
| Critical value $t_{29,\,0.95}$ | 1.699 |
| $p$-value | 0.407 |
| 95% CI for $\mu$ | (8.70%, 12.14%) |

*Effect of the outlier.* I removed the 26.30% loan and reran the same $t$ test in R with `t.test()`. That gives $n = 29$, $\bar{x} = 9.87\%$, $s = 3.54\%$ and $t = -0.53$. The conclusion does not change, and without the outlier the sample mean even falls *below* the benchmark.

## 6. Conclusion

**(A) Decision.** The test statistic $t_{obs} = 0.24$ is well below the critical value 1.699, and $p = 0.41 > \alpha = 0.05$. So I **fail to reject** $H_0$.

**(B) Interpretation.** My sample does **not** give convincing evidence that 3-year LendingClub borrowers in early 2018 paid more, on average, than the 10.22% banks were charging for personal loans.

- **Rates vary a lot.** The sample's average (10.42%) was only 0.2 points above the benchmark. Rates vary so much from loan to loan (SD about 4.6 points) that a difference that small could easily come from chance alone.
- **Plausible values.** The 95% confidence interval says the true average rate is plausibly anywhere from about **8.7% to 12.1%**. That range includes 10.22%.
- **This does not prove the rates are equal.** "Fail to reject" means my data could not tell the two apart, not that they are the same.
- **The size of difference I could detect.** To reject $H_0$, my sample mean would have had to be at least $10.22 + 1.699(0.8392) \approx 11.65\%$. So with 30 loans this study could only have picked up a difference of roughly 1.4 percentage points or more. A real but smaller difference could easily go undetected.

**(C) Limitations.**

- **The benchmark is not a perfect match.**
  - The bank rate is for **24-month** loans, and mine are 36-month loans.
  - More importantly, banks and LendingClub serve **different borrowers**. Banks turn away many riskier applicants, while LendingClub lends across grades A–G. So even if a difference were found, it would not show that LendingClub charges *the same borrower* more. The two groups of borrowers differ, which is a classic confounding problem.
- **Skew and small sample.** The right skew and the outlier make $n = 30$ adequate but not generous for a $t$ test. A small sample also gives the test little power to detect modest differences (see above).
- **The data file.** The 10,000 loans are a subset whose selection method is not documented. They cover only three months, and only loans that were actually issued. Applicants who were rejected, or who turned down a high-rate offer, are not in the data.

> **INSTRUCTOR NOTE: WHAT THE FULL DATASET SHOWS**
>
> Because you have the whole data file, you can check a sample's conclusion against it, which is rarely possible in real research. The mean rate over **all 6,970** 36-month loans in the file is **11.24%**, about one point *above* the benchmark. So in this file $H_0$ is actually false, and this sample's test missed it. That is a **Type II error**.
>
> It was not bad luck so much as low power. With a true difference of about 1 point and an SD of about 4.4 points, a sample of 30 has only about a **34%** chance of rejecting $H_0$. About 117 loans would be needed for an 80% chance. The student's conclusion is still correct as written: they said the evidence was not convincing, not that the rates were equal. That is exactly the right way to report a non-significant result.

# Appendix A: the sample

**Table A1.** The 30 loans drawn by the Simple Random Sample tool (seed 251), in the order they were drawn. All are 36-month loans. Only the interest rate is analyzed.

| Draw order | loan_id | Issued | Grade | Loan amount ($) | Interest rate (%) |
|---|---|---|---|---|---|
| 1 | 94 | Jan-2018 | A | 20,000 | 7.97 |
| 2 | 1981 | Mar-2018 | A | 13,000 | 7.97 |
| 3 | 594 | Feb-2018 | C | 9,525 | 15.05 |
| 4 | 1924 | Feb-2018 | A | 6,000 | 7.35 |
| 5 | 5976 | Feb-2018 | B | 40,000 | 9.93 |
| 6 | 2158 | Feb-2018 | A | 20,000 | 6.72 |
| 7 | 143 | Mar-2018 | C | 4,800 | 16.01 |
| 8 | 5381 | Jan-2018 | D | 1,000 | 17.09 |
| 9 | 3935 | Jan-2018 | B | 2,500 | 11.99 |
| 10 | 1424 | Mar-2018 | C | 5,525 | 16.01 |
| 11 | 7377 | Mar-2018 | A | 7,600 | 7.96 |
| 12 | 4470 | Jan-2018 | B | 3,600 | 11.99 |
| 13 | 3609 | Feb-2018 | B | 34,425 | 10.41 |
| 14 | 151 | Feb-2018 | A | 13,000 | 5.32 |
| 15 | 3814 | Mar-2018 | C | 3,200 | 13.58 |
| 16 | 9729 | Jan-2018 | B | 4,000 | 11.99 |
| 17 | 5731 | Feb-2018 | B | 23,100 | 9.44 |
| 18 | 4641 | Mar-2018 | B | 8,900 | 9.92 |
| 19 | 2468 | Jan-2018 | A | 10,000 | 6.08 |

| Draw order | loan_id | Issued | Grade | Loan amount ($) | Interest rate (%) |
|---|---|---|---|---|---|
| 20 | 5821 | Jan-2018 | A | 22,000 | 5.32 |
| 21 | 1115 | Jan-2018 | A | 10,000 | 7.35 |
| 22 | 1727 | Feb-2018 | A | 20,000 | 7.96 |
| 23 | 9154 | Mar-2018 | A | 20,000 | 7.34 |
| 24 | 7390 | Feb-2018 | C | 12,000 | 16.02 |
| 25 | 8867 | Jan-2018 | B | 25,000 | 9.44 |
| 26 | 9129 | Feb-2018 | A | 8,000 | 6.08 |
| 27 | 3660 | Jan-2018 | A | 20,000 | 7.97 |
| 28 | 5429 | Jan-2018 | E | 30,150 | 26.30 |
| 29 | 983 | Jan-2018 | A | 24,725 | 6.08 |
| 30 | 1470 | Mar-2018 | B | 15,000 | 9.92 |

# AI Use Disclosure

> **WHAT A DISCLOSURE LOOKS LIKE**
>
> The project requires you to disclose any AI use. This section is an **illustration of the format**. It is here so you can see what a complete disclosure contains. Write yours about what *you* actually did.

- **Service provider:** Anthropic (Claude).
- **Prompts I used:**
  1. "Where can I find the average interest rate banks charge on personal loans, month by month?"
  2. "My t test failed to reject. How do I explain that in plain language without saying the two means are equal?"
- **How I used it.**
  - Prompt 1 pointed me to FRED. I found the series, checked it on the Federal Reserve's page, and chose the February 2018 value myself.
  - Prompt 2 helped me word Part 6(B). I rewrote its suggestion in my own words and checked it against my lecture notes on significance tests.
  - All calculations were done with the stat-tools site and my own hand calculations in Part 5(C). The one exception is the outlier check, which I ran in R with `t.test()`. No AI-generated numbers, figures or data appear in this report.
- **Citation:** Anthropic. (2026). *Claude* [Large language model]. https://claude.ai. Responses to prompts 1 and 2 above, accessed September 2026.